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ESE323

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Final Paper - Birthday Countdown Timer

Project Description:

This project is designed to count down to a user's set birthday with nine 7-segment LEDs. Upon booting up, the device prompts the user to input today's date before prompting the user to input their date of birth. The user inputs these dates using three tactile switches, including one set/reset button and two buttons that act as up and down buttons. Upon reaching the user's set date of birth, the device plays the 'Happy Birthday' song and begins the countdown again upon the conclusion of the song.

Circuit Description:

Three images of the design's circuit layout are shown below. The circuit utilizes an ESP32-DEVKITC-32E as a microcontroller, which also works as a power source for the circuit and its components. A DS3231MZ+TRL real time clock is used to collect today's current date and operate the timer.

With nine XZFMDK07C common-cathode 7-segment LEDs necessary for the design, a MAX6951CEE+ display driver is used to drive eight of the nine 7-segment LEDs. This display driver minimizes the connections to the microcontroller and simplifies some of the programming for the device. As the MAX6951CEE+ can only control up to eight 7-segment LEDs, the ninth and final LED is connected directly to our ESP32-DEVKITC-32E microcontroller.

Also connected directly to our microcontroller is our SMTB-0927-TW-R buzzer, programmed to play the 'Happy Birthday' song upon reaching 000:00:00:00 on the device. Both the buzzer and our common-cathode 7-segment LED require current limiting resistors (R7-R15). Note that the 7-segment LEDs connected to the MAX6951CEE+ display driver do not require current limiting resistors.

Capacitors of varying magnitude are used throughout the circuit to minimize noise and ensure component functionality.

Finally, our three B3S-1000 tactile switches are connected directly to the microcontroller with an RC filter to help with debouncing. These switches are each active-low, utilizing a pull-up resistor. The circuit also includes four grounded mounting holes as well as various test points connected to important signals such as CLK, GND, and VDD.

Design Images/Diagrams:

Attached below are a series of images and diagrams relating to the project design. The attachments are organized in the following sequential order:

1. Bill of materials
2. Schematic diagram
3. Board layout (all metal layers)
4. Board layout (visible from the top)
5. Board layout (visible from the bottom)
6. Unpopulated board photograph
7. Built board photograph
8. 3D enclosure bottom case view
9. 3D enclosure top case view
10. 3D enclosure top case view #2
11. 3D enclosure showing the circuit board

Following these photos is a description of the results of the project, as well as a description of what I would do differently next time with the project.

Results:

The result of this project was largely successful and the device operates mostly as intended. On power up, the device displays ‘year 2024’ with the 7-segment displays, and the user must set what year it currently is with the up and down arrow buttons (from a range of 2024-2099). Upon pressing the ‘set/reset’ button, the device displays ‘month 01’ and so on until the user finishes entering the full date and time, down to the second.

After the date is set on the device, the device displays ‘birthday’, prompting the user to enter their date of birth into the device. The user will be prompted to enter an hour, minute, and second for the timer to go off on their birthday. If the user wishes for the device to go off at midnight, these values can simply be left as their default (0). Upon entering their date of birth with the tactile switches, the device enters ‘countdown’ mode, in which the display shows the time in days:hours:minutes:seconds until the user's set date of birth.

Upon reaching 000:00:00:00 on the timer, the device does in fact play the ‘Happy Birthday’ song with the buzzer. Shortly after the conclusion of the song, the countdown begins again at 365:00:00:00. The device does account for leap years and will not allow the user to input a date that does not exist (i.e. September 31st).

The device also has functionality to change the date of birth after it has already been set by holding down the set/reset button. This takes the user back to the date of birth setup, again prompting for month, day, minute, and second values. Unfortunately, this function does not operate properly, and the user cannot actually change the values from what they have been set to originally. This stems from a flaw in the device’s programming, not the hardware itself.

Upon powering off, the device has to be set with the current date and time again as well as the user’s birthday.

What I would do differently:

If I were to make this device knowing what I know now, I would almost certainly have not utilized a real-time clock component in this device. With the ESP32-DEVKITC-32E's Wi-Fi capabilities, it would have been much more efficient, accurate, and cost effective for the device to collect the current time over Wi-Fi, rather than having the user set the current date and time upon each power cycle.

I would have also programmed the device to store the user's date of birth and remember the date of birth even when powered off. I believe my hardware has the capability to store this information, but I was unable to figure out how to program it in this manner. I could have also resolved this issue through hardware by utilizing the V_{BAT} pin of the DS3231MZ+TRL RTC, allowing the RTC to run even while the ESP32-DEVKITC-32E is powered off. This solution, however, would introduce further complications with the introduction of a battery into the circuit design.

One final change I would have made to the design is that I would have laid the components out on the circuit board in a more efficient manner. I had tried to organize my components on the board such that connected components would be closer to each other in order to minimize trace length. I did not do a great job of this, and upon running the autorouter in Fusion360 I had far too many vias in my design, about 86. Furthermore these vias were quite close to some of my component leads, introducing some difficulties with soldering and avoiding short circuits.